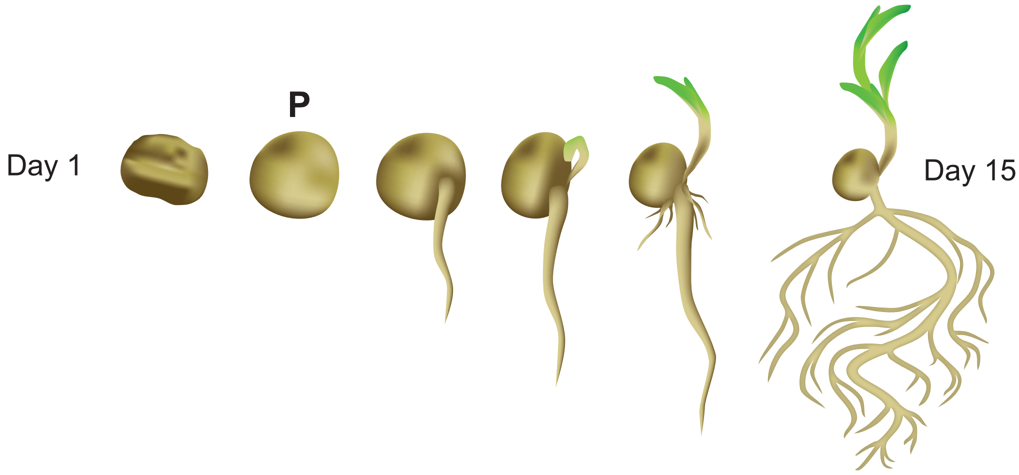
Examiner only

2. Germination involves the rapid onset of biochemical activity and growth of a seedling until the plant can carry out photosynthesis and become independent of the food stores contained in the cotyledons. **Image 2.1** shows some stages of a germinating pea plant over a 15-day period.

Image 2.1



(a) (i) The germinating pea seed labelled **P** in **Image 2.1** was cut open. This is shown in **Image 2.2.**

Image 2.2

Add three labelled lines to identify the **cotyledon, plumule and radicle** on **Image 2.2**. [2]

(ii) State **two** reasons why it is important that the radicle emerges prior to the plumule. [2]

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(b) Briefly describe the stages of germination in a **non-endospermic** seed such as the pea. [3]

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Question continued overleaf

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Gibberellin is required for maize kernel (seed) germination. It is blocked by another plant hormone, abscisic acid (ABA) whilst the kernel is maturing on the cob. Mutant kernels which cannot produce ABA germinate whilst still on the cob. This is shown in **Image 2.3**.

Image 2.3

(c) With reference to the function of gibberellin, explain why the mutant kernels germinate causing the appearance of the corn cob in **Image 2.3**. [3]

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(d) Field trials were carried out to investigate the effect of increasing soil temperature on the average number of wheat plants that grow. The results are shown in **Table 2.4**. In each field, the wheat grains were planted at a depth of 4 cm at a rate of 100 kg per hectare and the temperature was measured in each field at a depth of 5 cm.

Table 2.4

Mean maximum soil temperature /°C	Number of mature wheat plants per m^2
20.2	315.3
33.2	256.7
42.2	89.8

(i) One simple conclusion from this data is that higher soil temperatures reduce the number of mature wheat plants per square metre.
State **four** other factors that could affect the growth and so your confidence in this conclusion. [2]

I. ....

II. ....

III. ....

IV. ....

(ii) Assuming that this was a valid conclusion, state why the results of this research are of serious significance to worldwide wheat production. [3]

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